

# Image Analysis & Hex Dumps

## IT110 Lab 4 - Section B

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February 5, 2026

## Section B: Objectives

In this section, you will:

- Navigate the file system to locate image resources.
- Differentiate between uncompressed (TIFF) and compressed (JPEG) formats.
- Identify file signatures ("Magic Numbers") in Hexadecimal.
- Observe how raw pixel data looks compared to compressed data.

# Step 0: Navigation

First, you need to move from the text directory to the image directory.

## 1. Go back one level:

```
cd ..
```

## 2. Enter the image directory:

```
cd text_file
```

## 3. List the files:

```
ls -alth
```

# Step 0: Navigation

First, you need to move from the text directory to the image directory.

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## 3. List the files:

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```

**Check:** Do you see the .jpeg and .tiff files?

# Step 1: Inspecting File Types

Use the `file` command to see what kind of data you are dealing with.

```
file 16x16_16ppi_no_compression_gryascale.jpeg
file 16x16_16ppi_no_compression_grayscale.tiff
```

## Understanding the Difference

- **TIFF:** An uncompressed format. Every pixel value is represented directly as a hexadecimal integer.
- **JPEG:** A lossy compression format. It uses the *Discrete Cosine Transform* (DCT) to save space.

## Step 2: Dumping the JPEG

Let's look at the raw code of the JPEG file.

```
hexdump -vC 16x16_16ppi_no_compression_gryascale.jpeg
```

### Command Breakdown:

- `hexdump`: The command to "dump" the file content.
- `-v`: **Verbose**. Shows the entire output (doesn't hide repeated lines).
- `-C`: **Canonical**. Shows Hex on the left, ASCII on the right.

## Step 3: Magic Numbers (Signatures)

Every file type has a "Signature" or "Magic Number" that tells the Operating System what it is.

**Task:** Look at the very beginning of your hexdump output.

### JPEG Signature

You should see FF D8.

This hexadecimal string confirms the file is a JPEG.

## Step 6: File Terminators

Just as files have a start code, they have an end code.

**Task:** Run the hexdump command again and look at the very end of the output.

```
hexdump -vC 16x16_16ppi_no_compression_gryascale.jpeg
```



## Step 6: File Terminators

Just as files have a start code, they have an end code.

**Task:** Run the hexdump command again and look at the very end of the output.

```
hexdump -vC 16x16_16ppi_no_compression_gryascale.jpeg
```

### JPEG Terminator

You should see FF D9.

This indicates the End of File (EOF) for a JPEG.

## Step 4: Dumping the TIFF

Now, let's look at the uncompressed TIFF file. We will save this output to a text file to study it.

```
xxd -p 16x16_16ppi_no_compression_grayscale.tiff > hexdump_TIFF.txt
```

## Step 5: Analysis

Open the file in your text editor (or use cat). Scroll through it.

**Q: Do you see a pattern like this?**

00 00 00 00 ... FF FF ...

## Q: What is that pattern?

You are looking at an 8-bit grayscale image.

- 00 represents **Black**.
- FF (255 in decimal) represents **White**.

Because TIFF is **uncompressed**, you can actually "see" the image content (the pixels) directly in the hexadecimal code!

## Step 7: Reverting Hex to Image

Finally, let's try to convert a hex dump back into an image file.

### Command:

```
xxd -r -p hexdump_JPEG.txt output.jpeg
```

### Critical Question

Did you see the 00 00 FF FF pixel pattern in the JPEG hex dump?

## Step 7: Reverting Hex to Image

Finally, let's try to convert a hex dump back into an image file.

### Command:

```
xxd -r -p hexdump_JPEG.txt output.jpeg
```

### Critical Question

Did you see the 00 00 FF FF pixel pattern in the JPEG hex dump?

### Answer: No.

JPEG compression destroys the raw pixel map and replaces it with mathematical coefficients. You cannot "see" the image in the hex code of a compressed file.

# End of Section B